What is a scientific theory?

or, how might you best formalize a scientific theory, were you inclined to do so?

Hans Halvorson and Jim Weatherall

September 8, 2013

Philosophy of science meets Hegel

Syntactic view of theories: A theory is a collection of sentences in a formal language.

- Carnap
- Nagel

The semantic revolution: No, scientific theories aren't like that. A more accurate reconstruction of a scientific theory will present it as a family of set-theoretic structures.

- Putnam: Theories cannot be syntactic things.
- Suppes: A theory is a set-theoretic predicate.
- van Fraassen: A theory is a collection of models.

The semantic view of theories

Slogan:
A scientific theory should be identified with its class of models.

Innocuous?

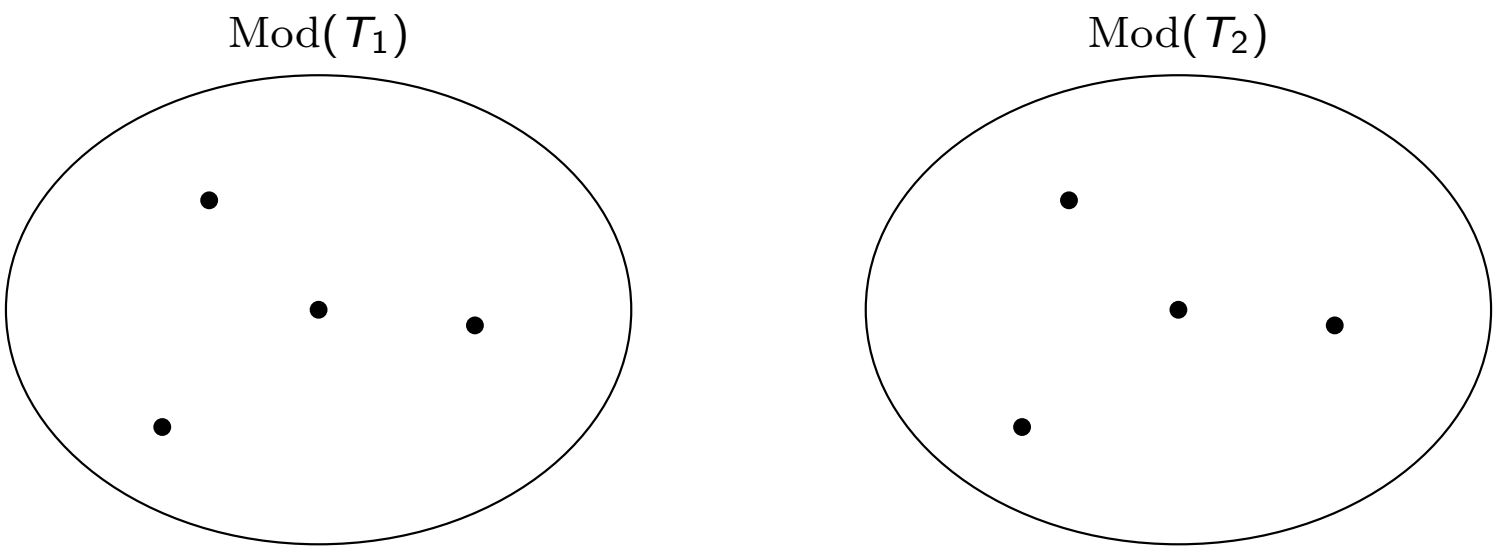
- Jill North: Lagrangian and Hamiltonian mechanics are inequivalent, because their models are not isomorphic
- To believe a theory is to believe that the WORLD is isomorphic to one of its models.

Otiose?

- van Fraassen: Empiricism didn't work in the syntactic framework; it does work in the semantic framework.
- John Bickle: Intertheoretic reduction didn't work in the syntactic framework; it does work in the semantic framework.

Motivating example I

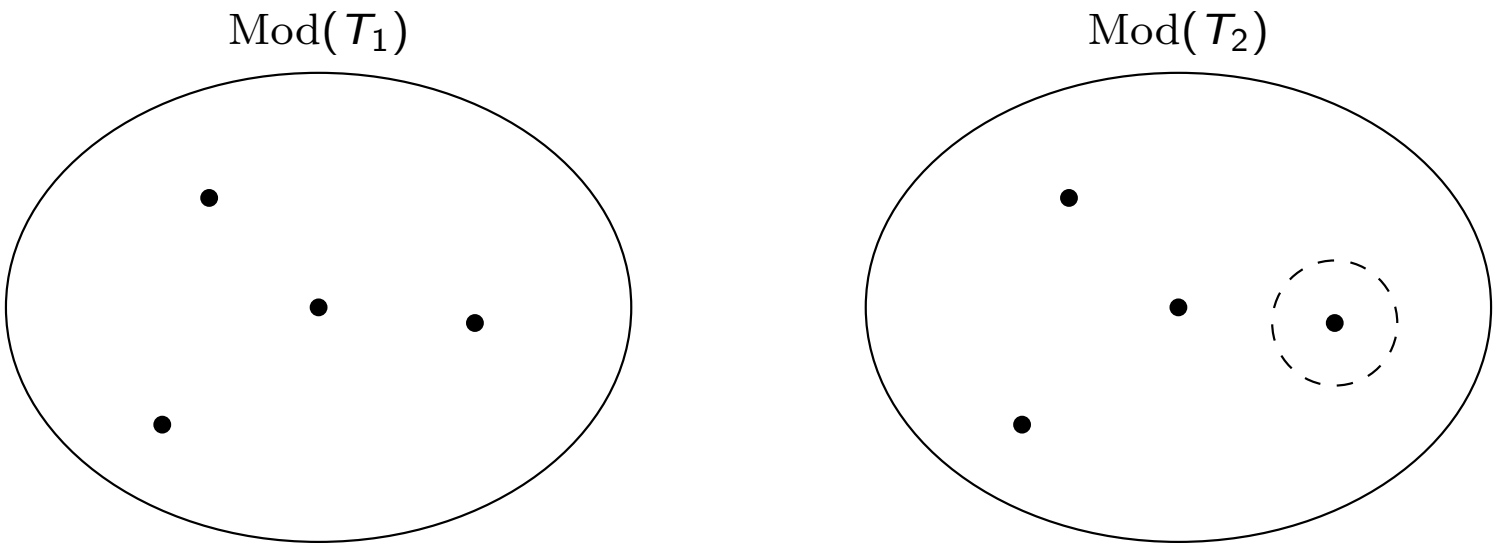
	Language	Axioms
T_1	$\{p_0, p_1, \dots\}$	$\emptyset$
T_2	$\{p_0, p_1, \dots\}$	$p_0 \vdash p_i$



Challenge: How to distinguish $\mathrm{Mod}(T_1)$ from $\mathrm{Mod}(T_2)$ when they are isomorphic sets.

Motivating example I

T_2 has a lonely model.



Moral: A theory doesn't just give us models, it also gives us a notion of closeness of models.

From a mathematical point of view

Question: Can a theory T be reconstructed from $\mathrm{Mod}(T)$?

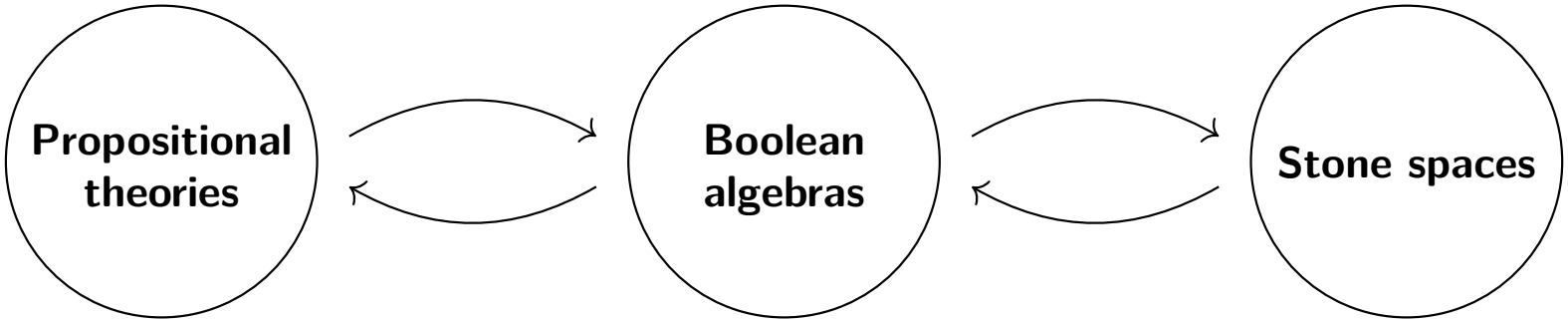
Clarification: What kind of mathematical object is $\mathrm{Mod}(T)$?

Answer: If $\mathrm{Mod}(T)$ is reduced to a bare set, then crucial information about T has been lost.

Example: which theory has as its typical model $\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}$?

Challenge: Where does $\mathrm{Mod}(T)$ live in the mathematical universe? What is the semantic content of a scientific theory?

Stone duality theorem



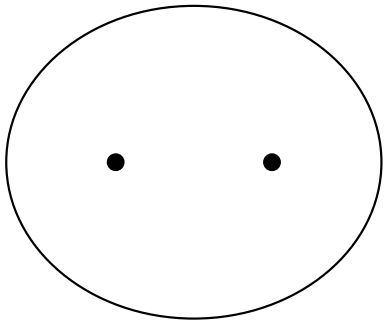
$\mathrm{Mod}(T)$ is a topological space; and the Lindenbaum algebra of T can be recovered from its topological structure.

Moral: A theory doesn't just give us models, it also gives us a notion of nearness of models.

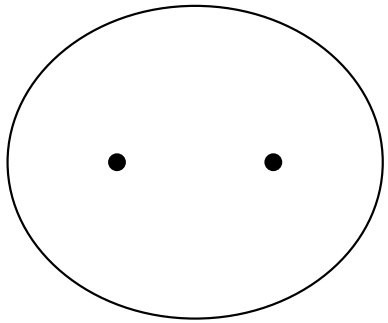
Motivating example II

	T_1	T_2
Language	$\emptyset$	$\{c\}$
Axioms	$\exists_{=2}$	$\exists_{=2}$

The unique model of T_1



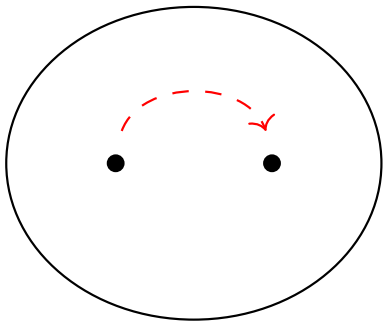
The unique model of T_2



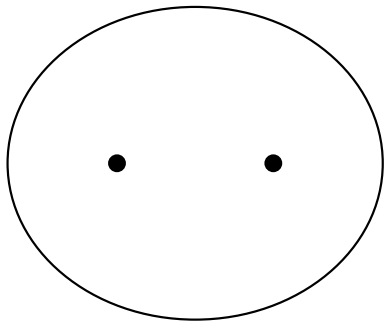
Challenge: How to distinguish $\mathrm{Mod}(T_1)$ from $\mathrm{Mod}(T_2)$ when each consists of a single isomorphism class?

Motivating example II

$\mathrm{aut}(M_1) = S_2$



$\mathrm{aut}(M_2) = (e)$



Moral: A theory doesn't just give us models, it also gives us arrows between models.

What do equivalent theories share in common?

Standard Account: T_1 and T_2 are theoretically equivalent just in case they have a common definitional extension. (Glymour 1970, "Theoretical realism and theoretical equivalence")

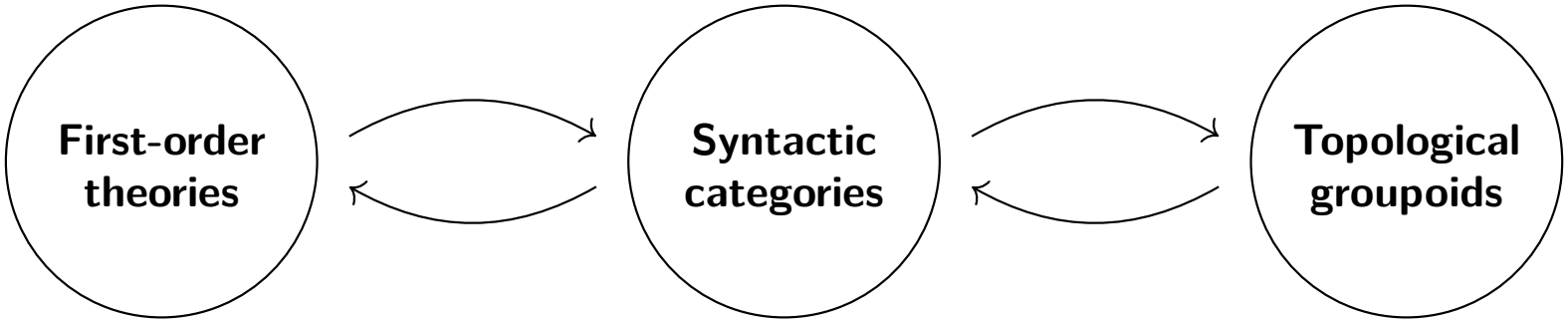
Fact: T' is a definitional extension of T just in case the forgetful functor $G : \mathrm{Mod}(T') \to \mathrm{Mod}(T)$ is:

- essentially surjective: each model of T can be expanded to a model of T'.
- full: each morphism between T models is also a morphism between T' models.
- faithful: automatic since G is the identity on arrows
- continuous in the logical topology: automatic

What do equivalent theories share in common?

Corollary: If T and T' are theoretically equivalent (in Glymour's sense), then $\mathrm{Mod}(T)$ and $\mathrm{Mod}(T')$ are isomorphic qua topological groupoids.

Awodey-Forssell duality theorem



$\mathrm{Mod}(T)$ is a topological groupoid; and the syntactic category of T can be recovered from its topological groupoid structure.

Moral: The semantic content of a theory includes relations between models, such as closeness of models and arrows between models.

Major morals

1. Syntactic and semantic notions are *dual* to each other.
   - No important philosophical claim turns on viewing theories semantically instead of syntactically.
2. The syntactic category of a theory is a hybrid semantic-syntactic object.
3. Language-dependency is not peculiar to a syntactic account; in all cases, we need a notion of equivalence of theories.

Complications and questions

- Why just isomorphisms? Why not all arrows?
  - Explanation via relations between models.
- Which topology?
  - For theories with a finite language, the logical topology is trivial.
  - What has the logical topology to do with other notions of similarity between models?
- Relations between theories

Repeating the mistakes of the past?

- The syntactic view accused of giving rise to fruitless internal questions. (see van Fraassen)
- The semantic view gives rise to questions about "representation" that are of no concern in scientific practice